

3D MODEL · SPEC CONVERSION

STEP → E3D Equipment

Heuristically recognizes NX STEP B-rep shapes and converts them to E3D primitives (CYLI / BOX / cylinder).
Handles nozzles / flanges / bolts.

Key Purpose

Why this program

1. Turn a vendor STEP model into an E3D Equipment macro

- Reads the NX B-rep solid assembly and writes a ready-to-run .mac that builds EQUI / SUBE / primitives
- **Effect:** Supplier 3D equipment is re-modelled in minutes instead of hours of manual primitive drawing, with no transcription error

2. Recognize shapes automatically as native primitives

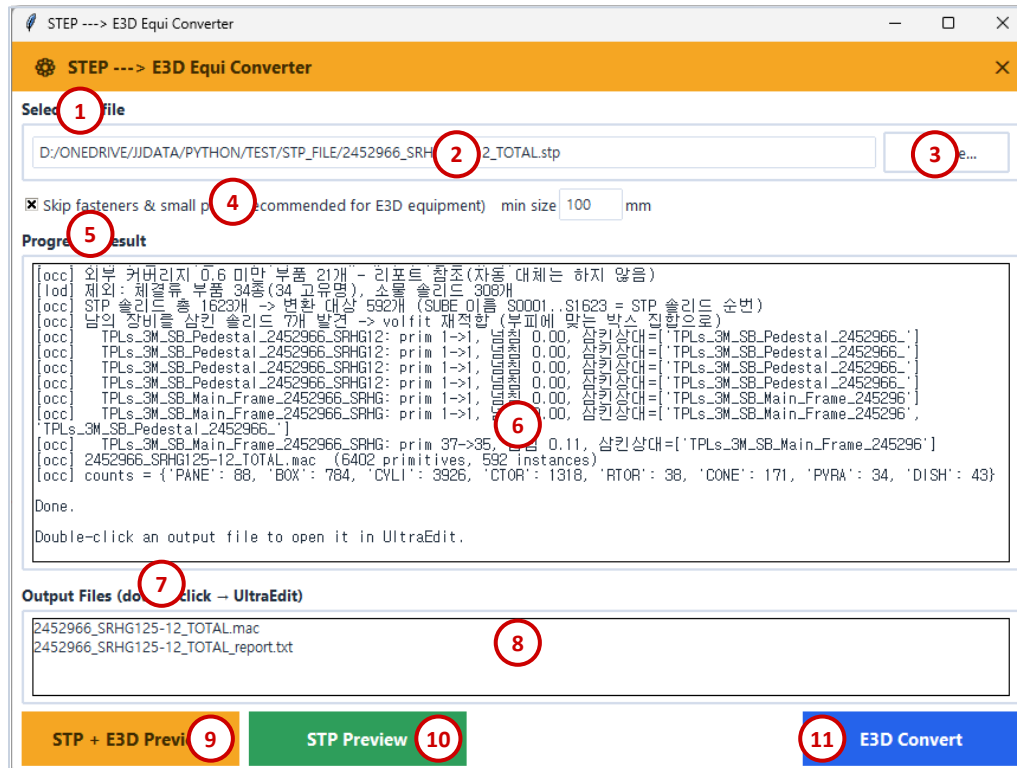
- Analyzes each solid's surfaces and maps them to CYLI / BOX / CONE / DISH / CTORUS — flanges, nozzles, bolts, elbows, structural sections
- **Effect:** The converted model uses real E3D primitives that can be edited and clash-checked, not a dumb mesh

3. Keep true world position per part

- Walks the STEP assembly tree and preserves each occurrence's absolute placement
- **Effect:** Every part lands where it belongs, so the equipment drops straight into the plant model without re-fitting

Required inputs : A STEP (.stp) B-rep solid assembly file

Main Window – Item Guide



Overview

What it does

1. Heuristically recognizes NX STEP B-rep shapes and converts them to E3D primitives (CYLI
2. BOX
3. Cylinder)
4. Handles nozzles
5. Flanges
6. Bolts

Category: 3D Model · Spec Conversion

Folder: E3D_STP_CONV

Main scripts: stp_to_e3d_equi.py

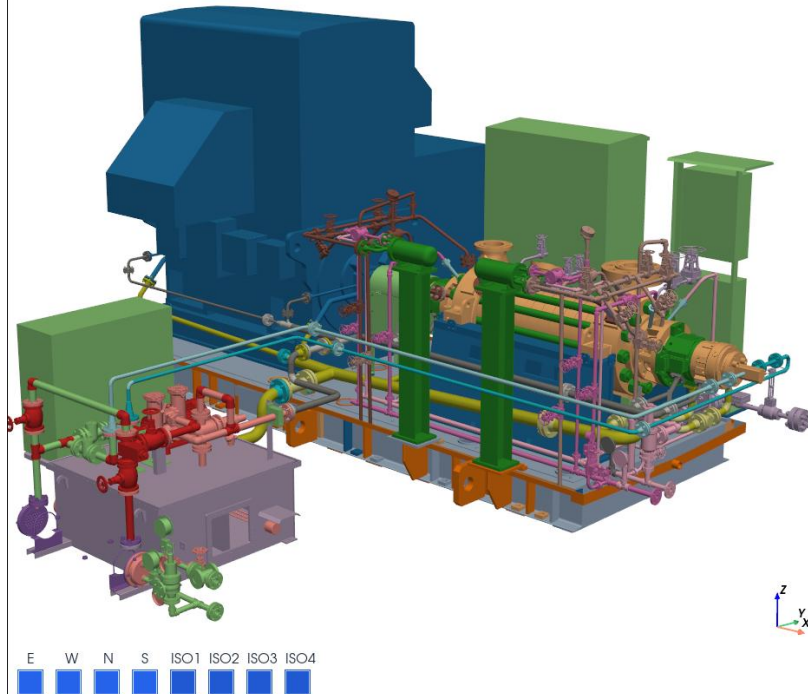
Scripts / Status: 4 scripts · Experimental

Tags: STEP · B-rep · E3D · primitive

Output Samples

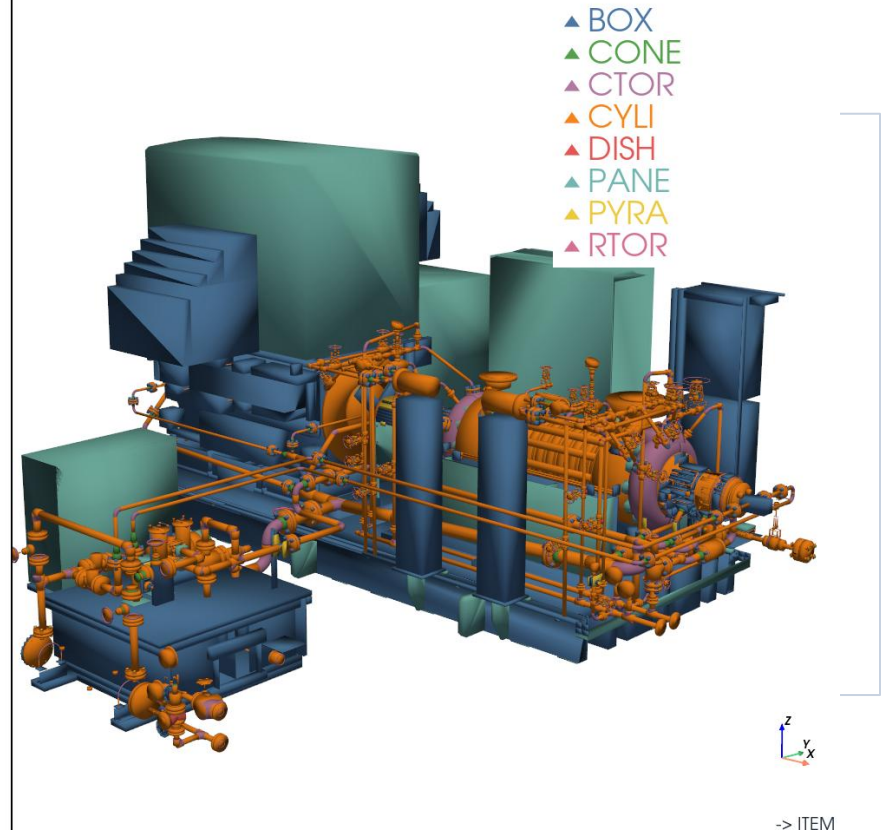
Input & Output Preview

STP
2,097,031



Input STP file Preview

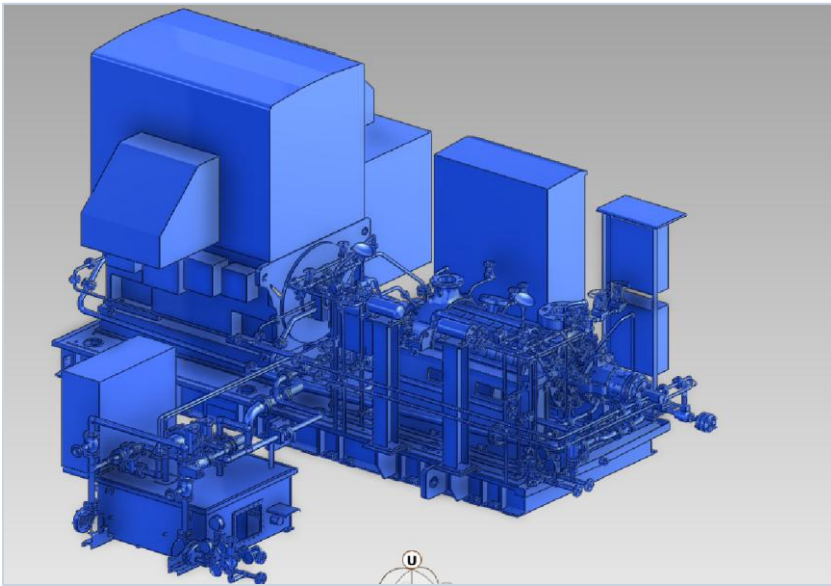
E3D 10,232
CYLI=5611 BOX=2425 CTOR=1506 CONE=393 PANE=127 PYRA=89



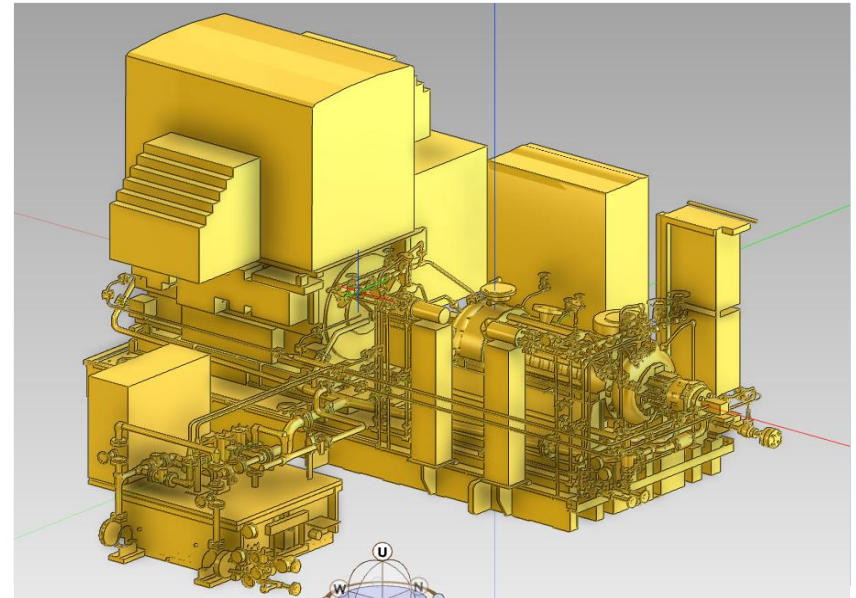
Converting Output Preview

Output Samples

Input & Output STP to E3D



Input STP

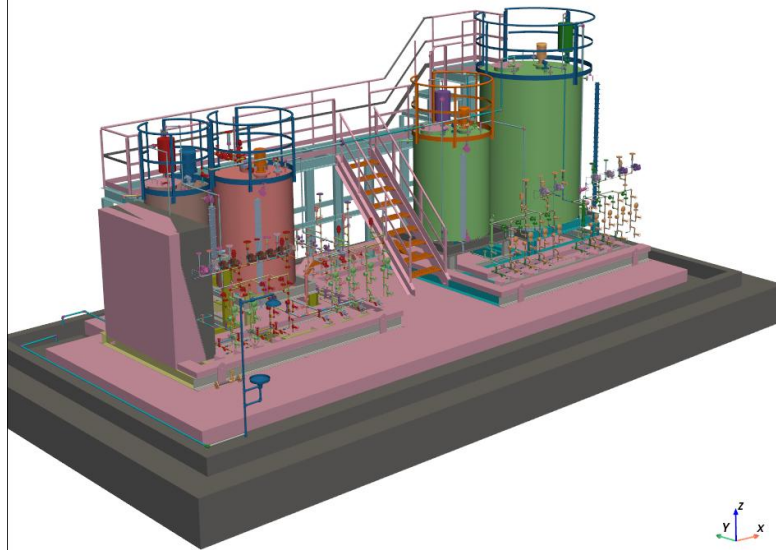


Output E3D Equipment

Output Samples

Input Output Preview

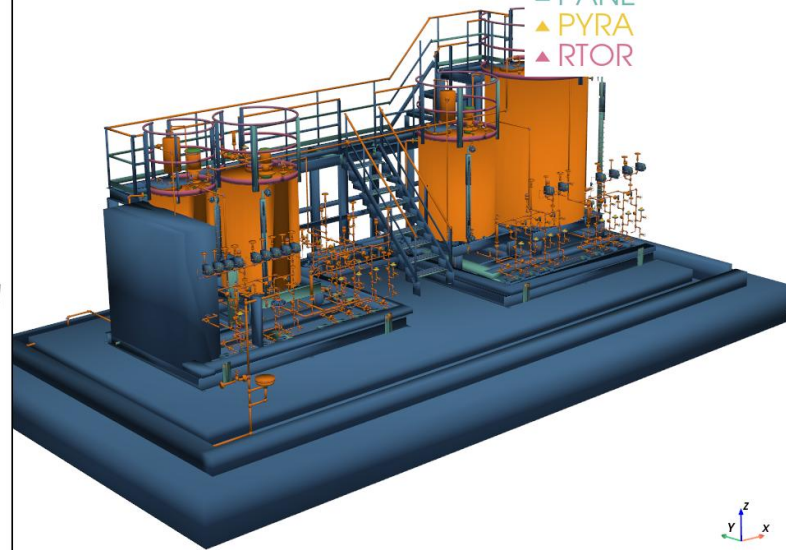
STP
4,499,289



E W N S ISO1 ISO2 ISO3 ISO4

Input STP file Preview

E3D 21,982
CYLI=10666 BOX=7069 CTOR=1510 CONE=1374 DISH=773 PYRA=401

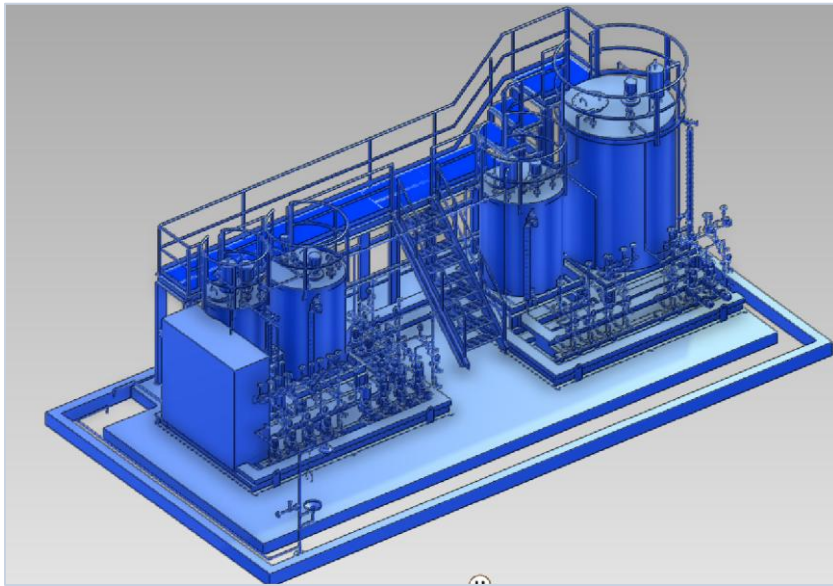


-> ITEM

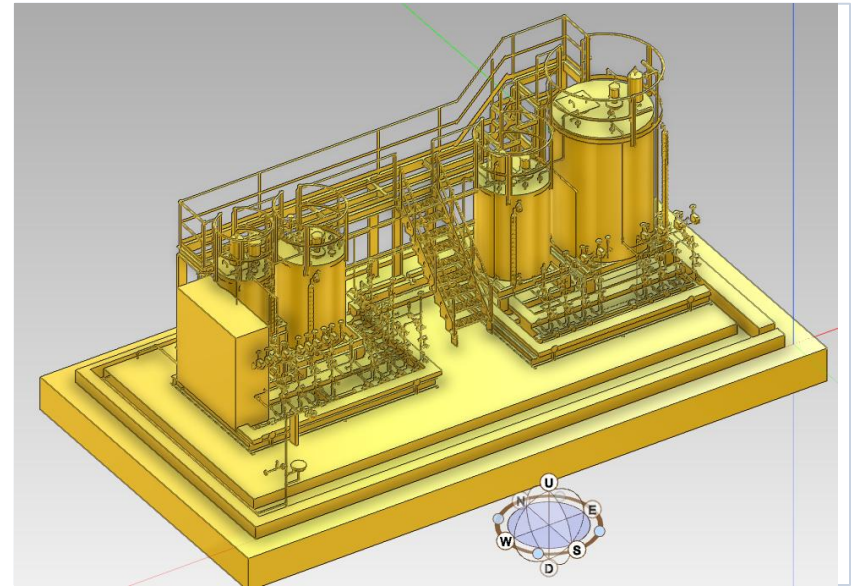
Ouput E3D Preview

Output Samples

Input Output Samples



Input STP



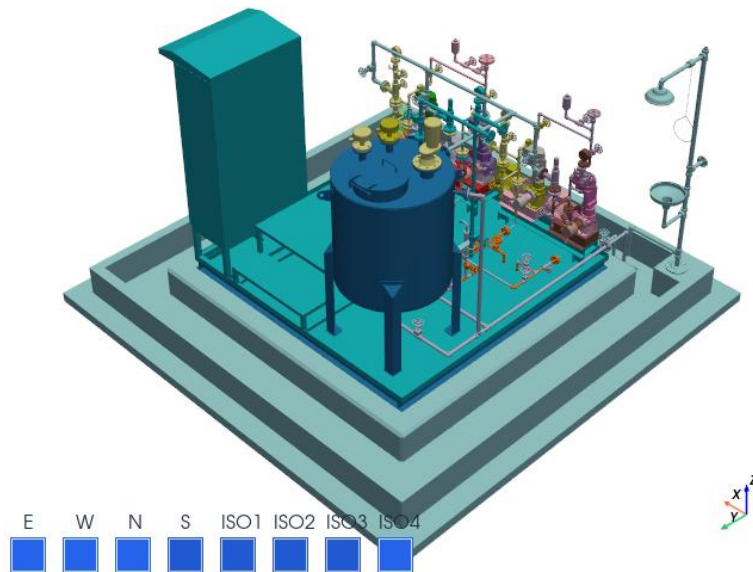
Output E3D Equipment

Output Samples

Input Output Preview

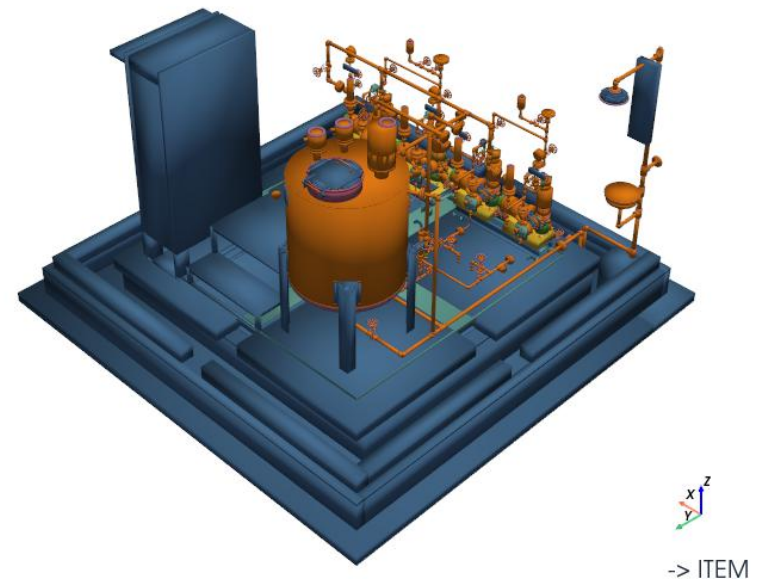
STP vs E3D ??Rabigh_2Phosphate_CDS.stp

STP
2,785,811



Input STP

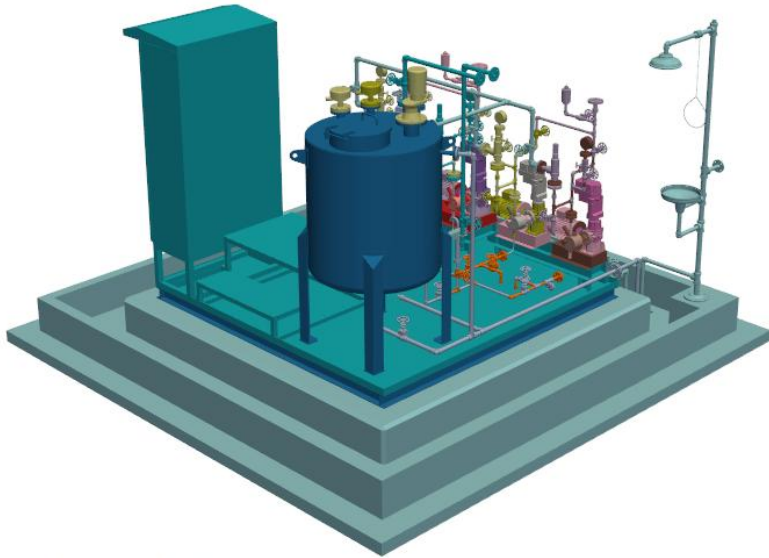
E3D 12.585
CYLI=5392 BOX=3647 CTOR=2201 CONE=626 DISH=505 PYRA=150



Output E3D Preview

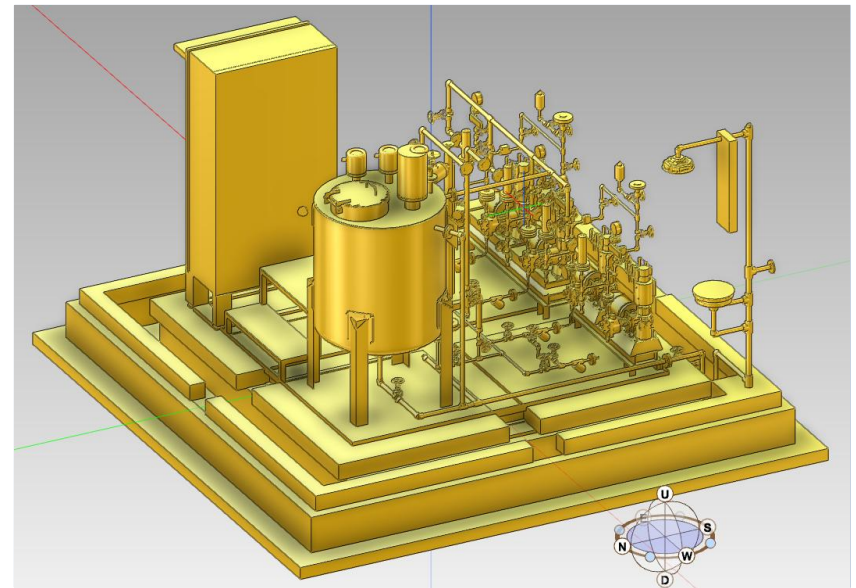
Output Samples

Input Output Sample



S ISO1 ISO2 ISO3 ISO4

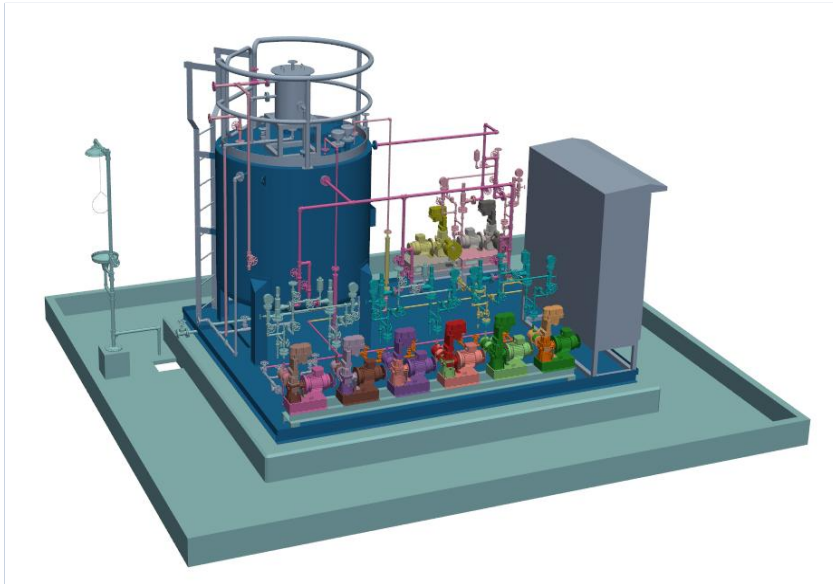
Input STP Preview



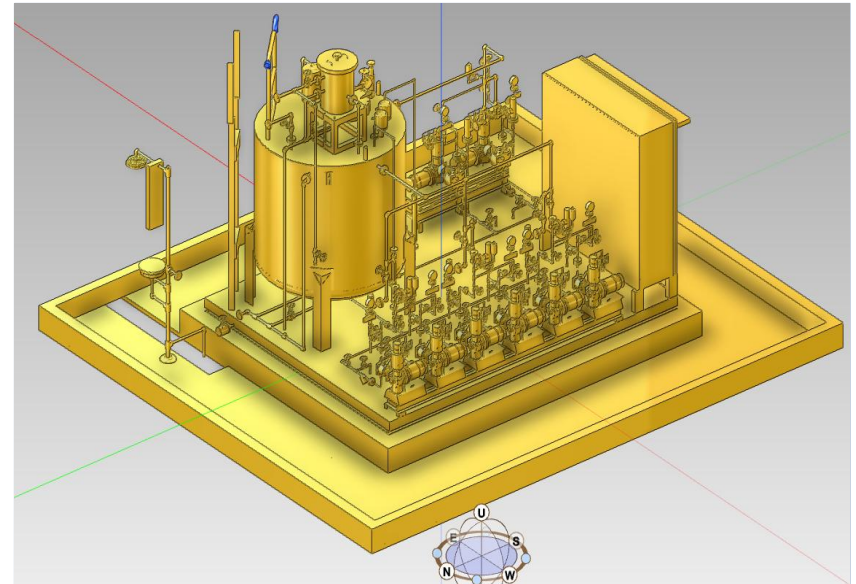
Output E3D Equipment

Output Samples

Input Output Samples



Input STD Preview



Output E3D Equipment

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: 3D Model · Spec Conversion.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: stp_to_e3d_equi.py.
- Current status: Experimental — behavior/outputs may still change.
- Output samples and the program window above reflect the current build.

Thank You

STEP → E3D Equipment ・ Plant Engineering Total Service System